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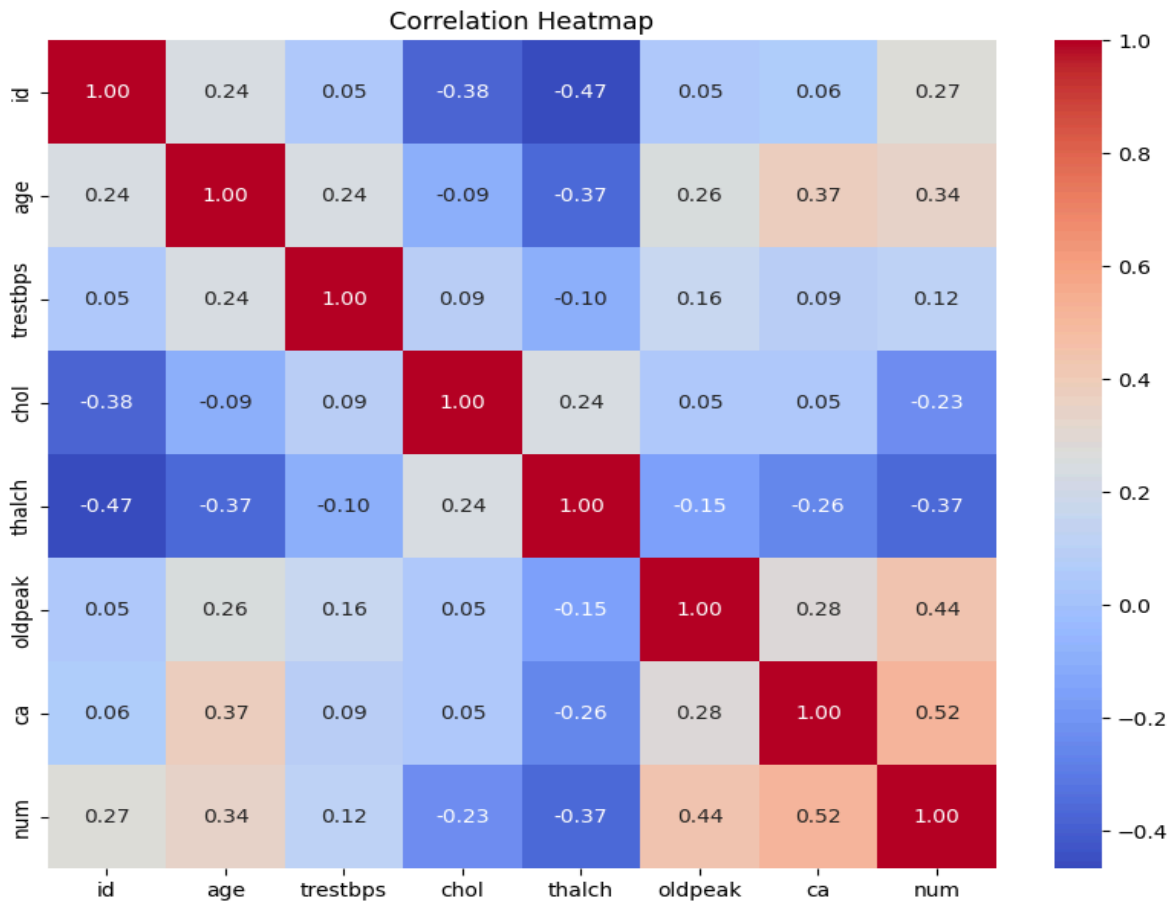
Introduction

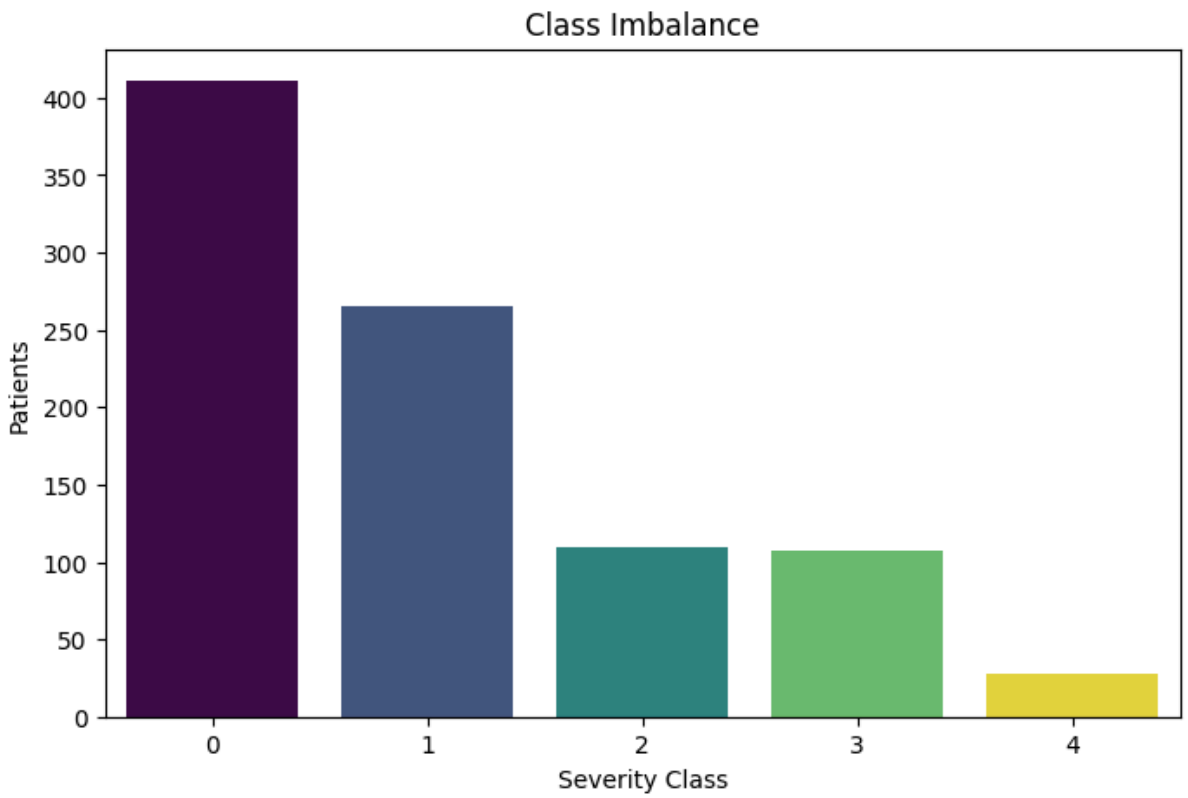
The aim of this project is to develop machine learning models capable of predicting the probability and intensity of heart disease in patients based on their medical data. Heart disease is one of the most common reasons for death in the world. By automating the diagnosis process using patient data such as age, cholesterol levels, and chest pain type this project aims to assist medical professionals to detect heart disease early, and reduce human error and improve patient health.

Dataset Description & EDA

The dataset contains 16 columns, 15 input features and 1 output feature. After dropping the not necessary id column, we have 14 input features to predict the target. This is a classification problem because the target variable num represents the intensity of heart disease in discrete categories 0 for no disease, and 1 through 4 for varying stages of intense heart disease, instead of a continuous numerical output. There are 920 datapoints in the dataset. The dataset contains both quantitative like age, trestbps, chol and categorical like sex, cp, fbs data. Here the categorical variables are needed to be encoded because machine learning models need numerical input to perform mathematical calculations. The text based categorical variables must be converted into numbers so the models can process them.



The heatmap shows that as some values of features, like oldpeak and age, increase, the risk of heart disease generally increases. Also as the values of some features like, thalch increases there is a low chance of heart disease



Here the bar chart shows that the dataset is heavily imbalanced. Here class 0 has the majority of the instances. Classes 1 through 4 have fewer data points, which makes it difficult to predict the exact severity level of the heart disease.

Dataset Pre-processing

During pre processing the id column was dropped because it had no valuable information and was an extra burden to train the model. There were some values that were missing in the dataset. Instead of removing their whole column for quantitative values the columns were filled using their medians and for categorical values the columns were filled with their mode instead of dropping the entire column. For categorical values one-hot encoding was applied to replace the texts with binary values since models can’t calculate without numbers. The numbers were also scaled using standard scaler because the scale of the numerical variables were very different because some numbers ranged between tens and others hundreds.

Dataset Splitting

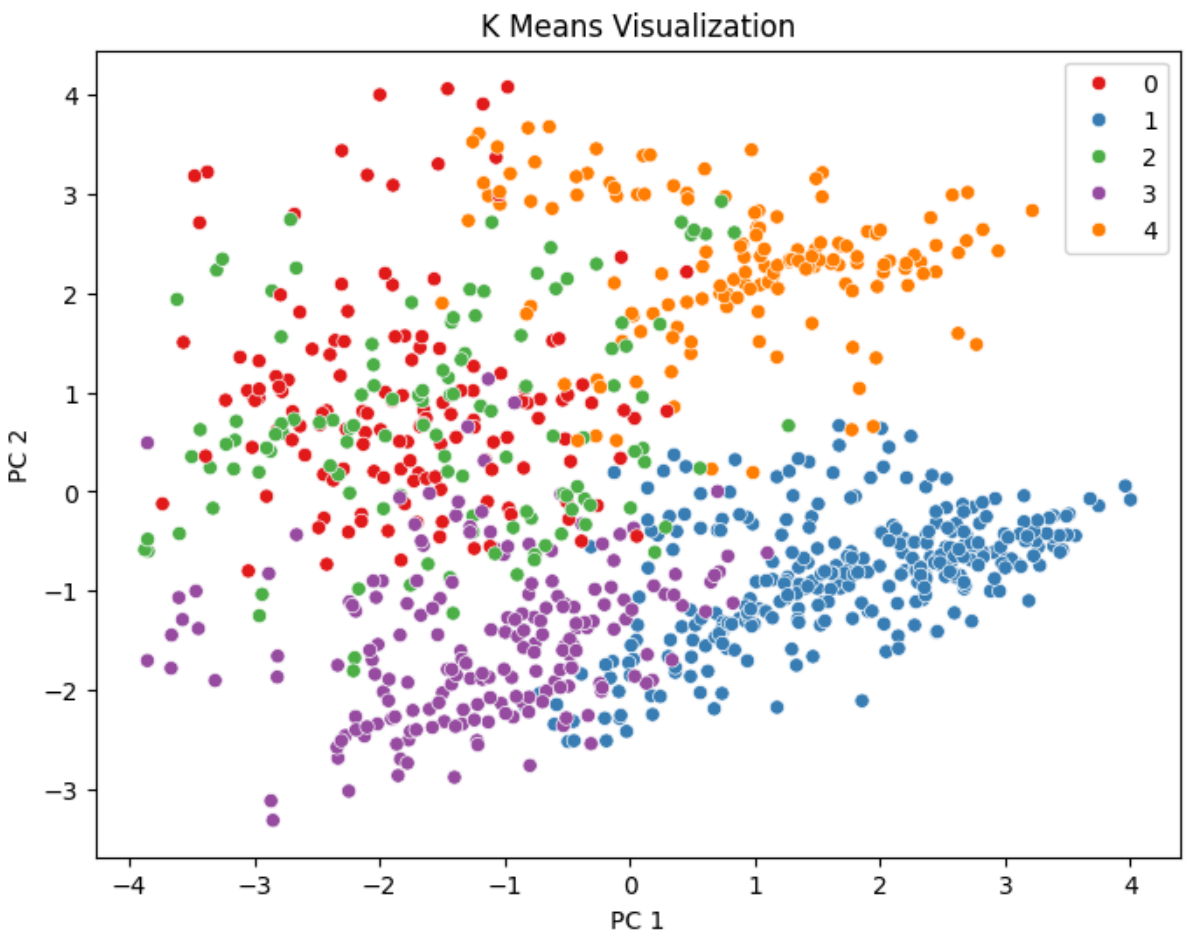
The dataset was randomly split with 80% data used to train and 20% data used to test.

Model Training & Testing

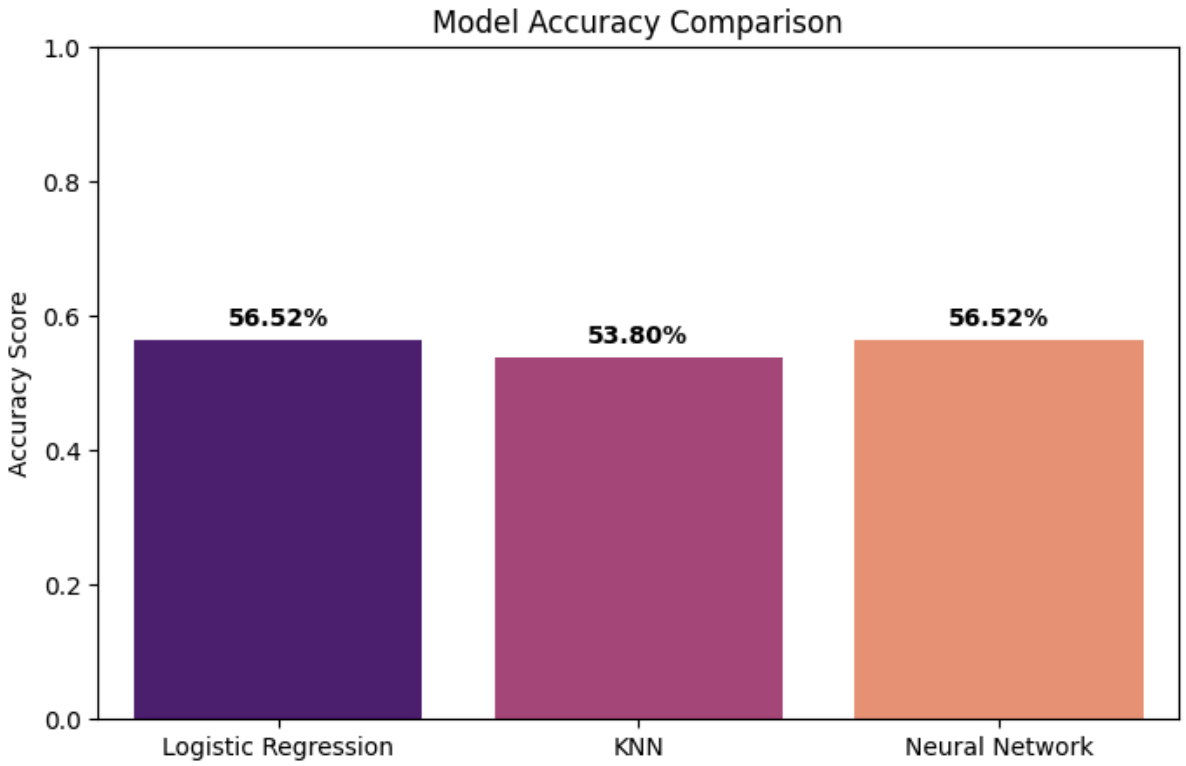
The following models were used

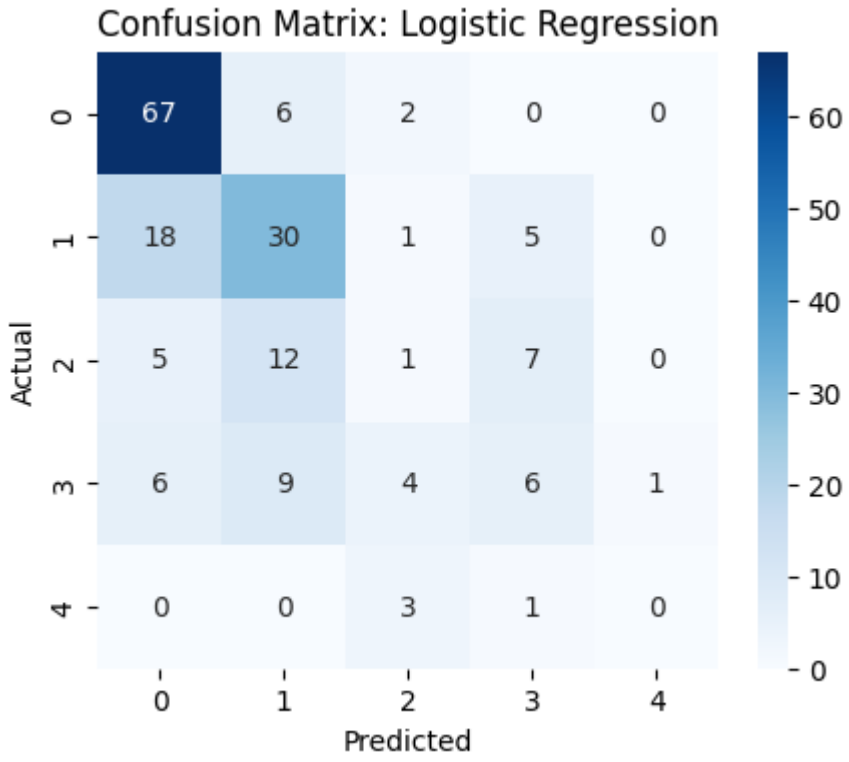
1. Logistic Regression
2. KNN
3. Neural Network

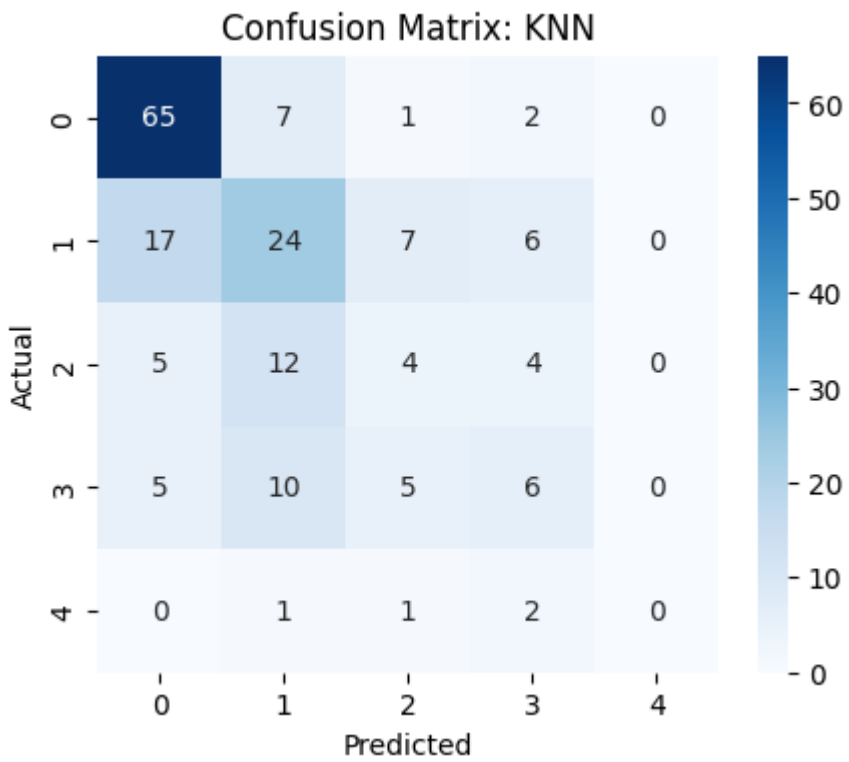
Also Kmeans cluster was applied with k = 5 to represent the 5 classes. Also PCA was used to reduce the dimension to represent all features on a 2D plot.

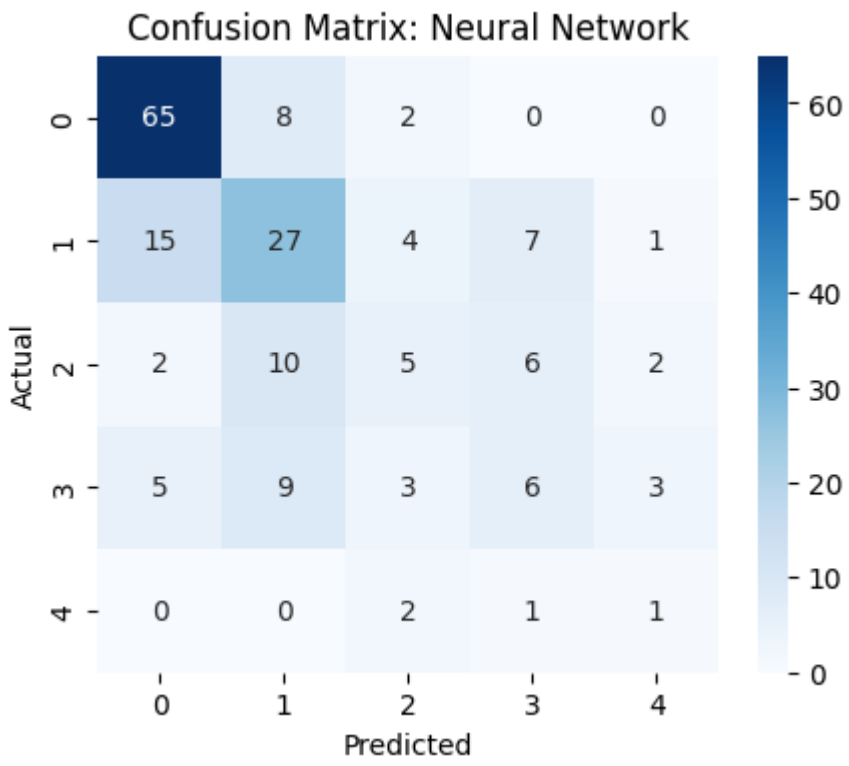


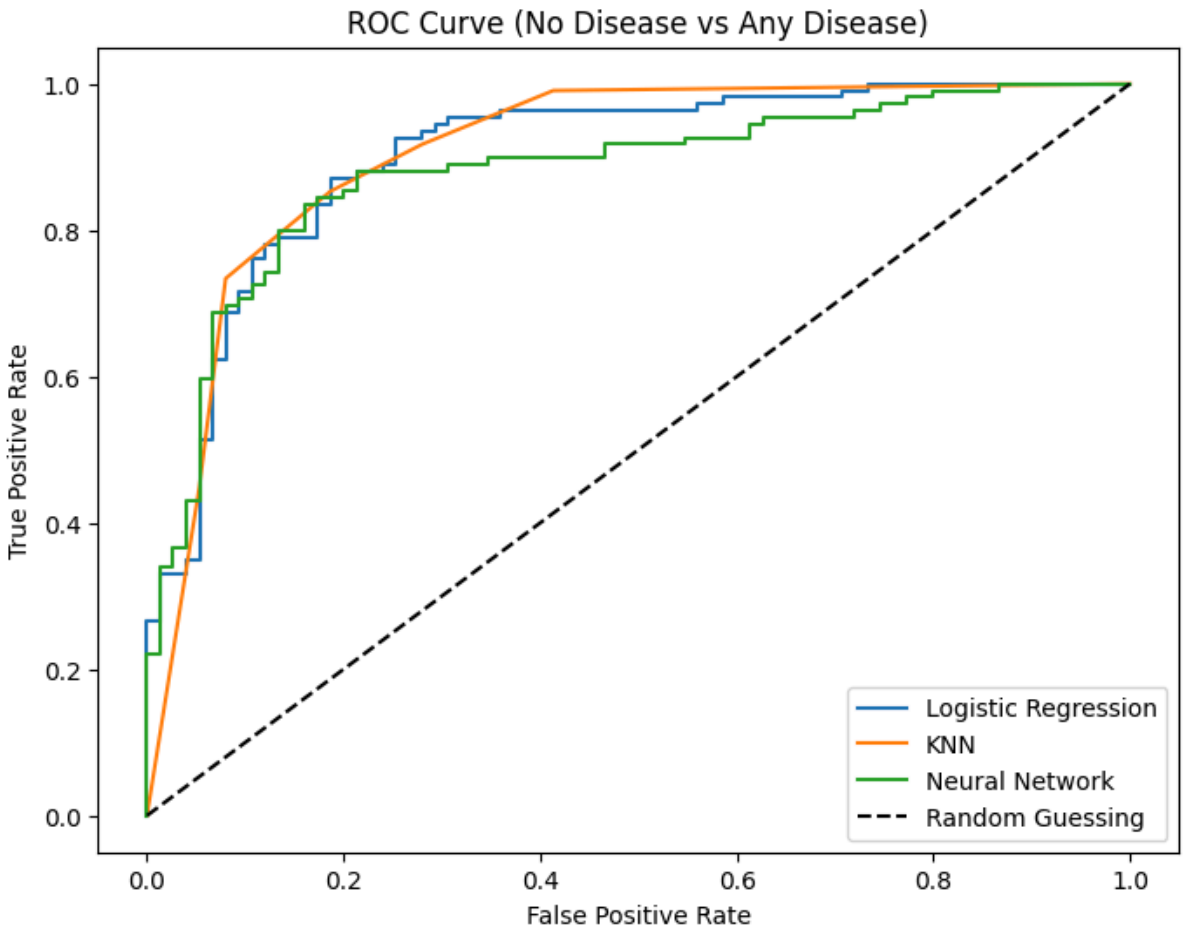
Model Comparison & Analysis




Confusion Matrix: Logistic Regression
Actual
Predicted
0
1
2
3
4
67 6 2 0 0
18 30 1 5 0
5 12 1 7 0
6 9 4 6 1
0 0 3 1 0
60
50
40
30
20
10
0


Confusion Matrix: KNN
Actual
Predicted
0
1
2
3
4
65 7 1 2 0
17 24 7 6 0
5 12 4 4 0
5 10 5 6 0
0 1 1 2 0
60
50
40
30
20
10
0


Confusion Matrix: Neural Network
Actual
Predicted
0
1
2
3
4
65 8 2 0 0
15 27 4 7 1
2 10 5 6 2
5 9 3 6 3
0 0 2 1 1
60
50
40
30
20
10
0


ROC Curve (No Disease vs Any Disease)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression
KNN
Neural Network
Random Guessing

Model Comparison & Analysis

The dataset was split into 80% training and 20% testing sets. Three models were analyzed: Logistic Regression, KNN, Neural Networks.

1. Logistic Regression

Accuracy: 56.52% | AUC Score: 0.8181

Class	Precision	Recall	F1-Score	Support
0	0.70	0.89	0.78	75
1	0.53	0.56	0.54	54
2	0.09	0.04	0.06	25
3	0.32	0.23	0.27	26
4	0.00	0.00	0.00	4
Accuracy			**0.57**	**184**
Macro Avg	0.33	0.34	0.33	184
Weighted Avg	0.50	0.57	0.52	184

2. KNN (K-Nearest Neighbors)

Accuracy: 53.80% | AUC Score: 0.6985

Class	Precision	Recall	F1-Score	Support
0	0.71	0.87	0.78	75
1	0.44	0.44	0.44	54
2	0.22	0.16	0.19	25
3	0.30	0.23	0.26	26
4	0.00	0.00	0.00	4
Accuracy			**0.54**	**184**
Macro Avg	0.33	0.34	0.33	184
Weighted Avg	0.49	0.54	0.51	184

3. Neural Network

Accuracy: 56.52% | AUC Score: 0.8119

Class	Precision	Recall	F1-Score	Support
0	0.75	0.87	0.80	75
1	0.50	0.50	0.50	54
2	0.31	0.20	0.24	25
3	0.30	0.23	0.26	26
4	0.14	0.25	0.18	4
Accuracy			**0.57**	**184**
Macro Avg	0.40	0.41	0.40	184
Weighted Avg	0.54	0.57	0.55	184

Conclusion

Based on the results, the models are generally effective at identifying patients without heart disease due to the high volume of training data for class 0. However, accuracy drops when attempting to classify the intensity levels classes 1-4. Here both logistic regression and neural network provided the best overall accuracy outperforming KNN. The results are heavily influenced by the class imbalance found in the EDA. Because the models have more examples of Class 0, they become biased toward predicting it. The challenges that was faced during making the project was mainly the preprocessing of the data where missing values had to fill up and scaling had to be used to have a consistent range of numbers. Also categorical values needed to be replaced to ensure models to train smoothly.